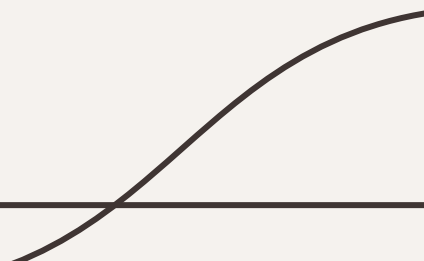


Maternal Mortality

Graph Out Loud

Tapheela Hood, Nicole Kopa, Sumaksharika Nainavarapu, Stuti Vyas



Agenda

01

Recap

02

Project Management
Overview

03

Methodology & Data
Source Description

04

Data
Visualizations

05

Challenges &
Learnings

06

Conclusion &
Key Takeaways



01

Recap

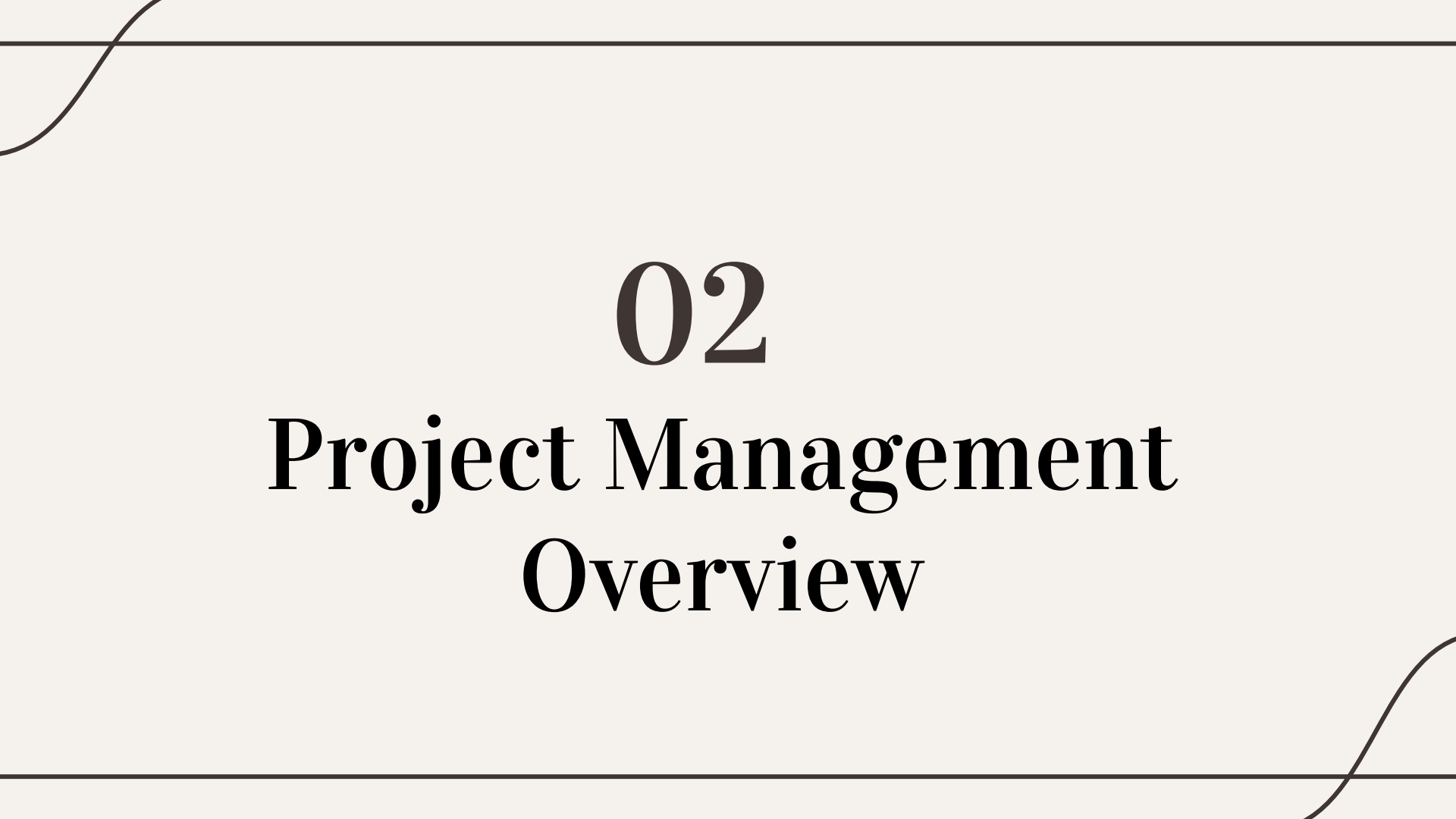
Research Question

In pregnant woman and new mothers in the United States **(P)**, how has the Covid-19 pandemic **(I)**, compared to the pre-pandemic period **(C)**, affect maternal mortality rates **(O)**, particularly in the years following the pandemic?

- What is the association between demographic and socioeconomic factors and the prevalence of maternal mortality?

Understanding Maternal Mortality

- According to WHO, maternal mortality refers to the **death of a woman during pregnancy, childbirth, or within 42 days of termination of the pregnancy**, from any cause related to or aggravated by the pregnancy or its management, **excluding those from accidental or incidental causes**.
- Most of the **complications that occur are preventable** through the health-care solutions (high-quality care).
 - Causes: severe bleeding, infections, high blood pressure (pre-eclampsia, eclampsia)
- Despite spending more than any other country in the world, the **United States has one of the highest** maternal mortality rates.
 - Key indicator of the quality of the healthcare systems, in context of maternal and reproductive health
- Based on the CDC numbers, Maternal mortality rates are expressed as the **maternal mortality ratio (MMR)**, number of maternal deaths per 100,000 live births within a given time frame



02

Project Management Overview

Overview

01

ClickUp: Project Manage

- Organize and manage projects 1 & 2
- Assign tasks, track progress and workflows
- Monitor status

02

Google Drive: Document Sharing

- Share documents, files, slide decks
- Collaborate seamlessly with team members
- Edit documents/slide decks simultaneously

03

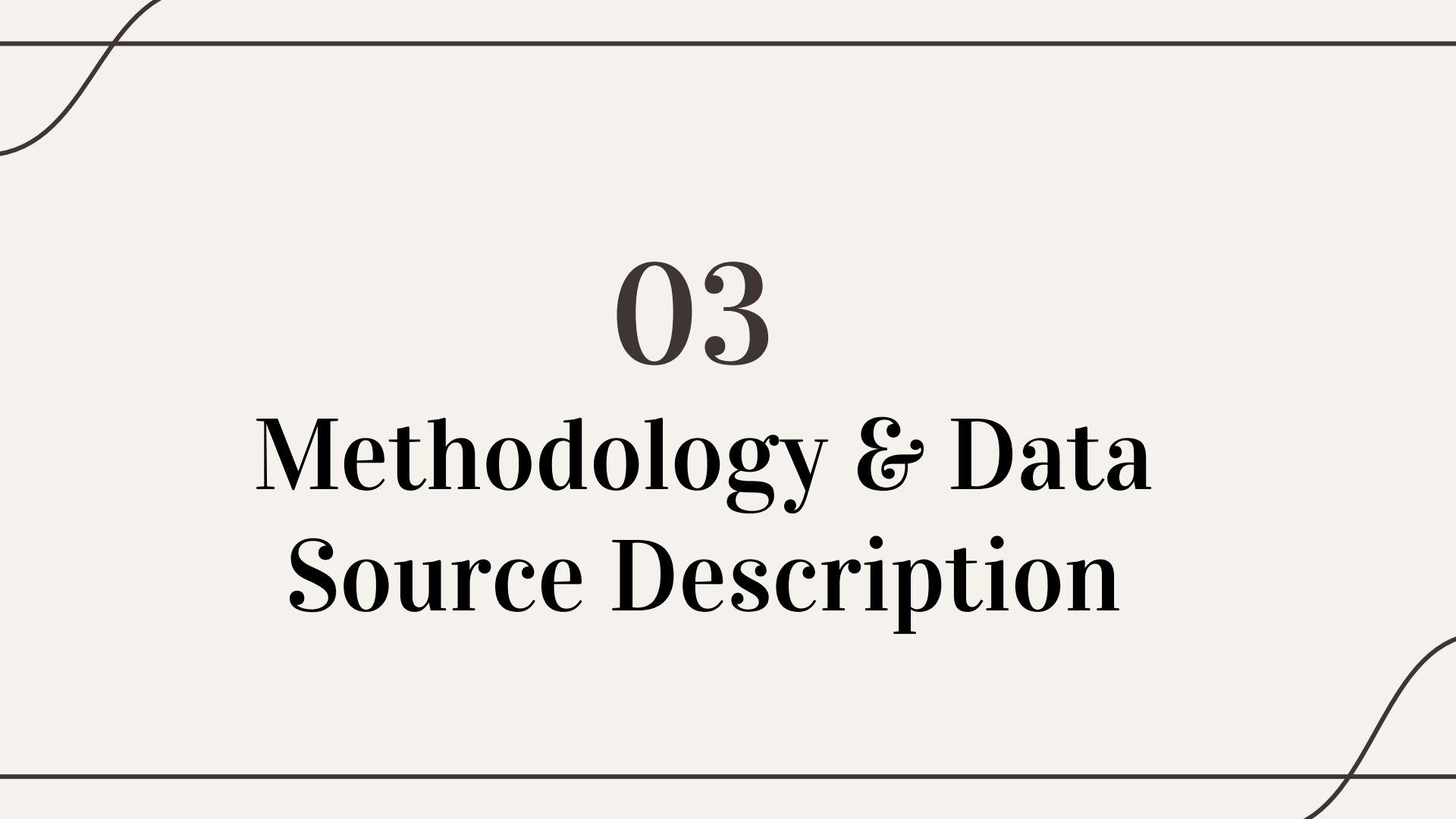
Zoom/Google Meet: Meetings & Collaboration

- Conduct virtual project meetings (Zoom for in class, Google Meet for outside of class)
- Host discussions & brainstorming sessions
- Prepare for presentations

04

iMessage: Continuous Project Discussions

- Facilitate real-time communication
- Share quick updates & feedback
- Discuss upcoming tasks



03

Methodology & Data Source Description

Methodology

Choosing a Topic

- Brainstorming on topic (maternal health, mental health)
- Initial analysis performed

Search on Datasets

- Efforts made to find reliable data sources/data sets
- Datasets identified: PRAMS, CDC VSRR Provisional Maternal Death Counts and Rates, CDC Pregnancy Mortality Surveillance Systems

Final Decision

- Pros and cons of datasets identified
 - Datasets chosen: PRAMS (2018-2022), CDC VSRR Provisional Maternal Death Counts and Rates (2019-2023) CDC Pregnancy Mortality Surveillance Systems (2017-2020)
-

Data Extractions & Merging

- **PRAMS: merging 2018 -2022 data** into excel with health indicators
- **Renamed variables** (health indicators)
- **Concatenated** some of the data points (health indicators)
- **Created** pivot tables

Pregnancy Risk Assessment Monitoring System (PRAMS), Maternal and Child Health (MCH) Indicators 2022														
Category	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13	Column 14
Health Indicator	Nutrition						Pre-pregnancy Weight						Overweight (BMI ≥ 25)	
	Multivitamin use 34 times a week during the month before pregnancy						Underweight (Body mass index [BMI] < 18.5 kg/m²)						Overweight (BMI ≥ 25)	
Site Name	N (Denominator)	N (Numerator)	Weighted %	Lower 95% Confidence Interval	Upper 95% Confidence Interval		N (Denominator)	N (Numerator)	Weighted %	Lower 95% Confidence Interval	Upper 95% Confidence Interval		N (Denominator)	N (Numerator)
Alabama	11221	1088	9.7	8.5	10.9		2051	182	8.9	7.6	10.2		2051	365
Alaska	261	289	36.3	33.4	39.2		251	16	6.4	3.6	9.0		251	27
Arizona	1221	80	6.5	5.1	8.0		889	31	3.5	2.6	4.4		889	21
Arkansas	1221	612	50.1	47.7	52.5		1247	80	6.4	5.4	7.4		1247	191
California	1221	1096	90.1	89.1	91.1		1214	49	4.0	3.4	4.6		1214	7
Colorado	1221	296	24.3	22.1	26.5		811	24	2.9	2.1	3.8		811	24
Connecticut	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Delaware	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
District of Columbia	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Florida	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Georgia	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Hawaii	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Idaho	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Illinois	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Indiana	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Iowa	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Kansas	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Kentucky	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Louisiana	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Maine	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Maryland	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Massachusetts	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Michigan	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Minnesota	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Mississippi	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Missouri	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Montana	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Nebraska	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Nevada	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
New Hampshire	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
New Jersey	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
New Mexico	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
New York State	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
North Carolina	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
North Dakota	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Ohio	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Oklahoma	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Oregon	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Pennsylvania	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Rhode Island	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
South Carolina	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
South Dakota	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Tennessee	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Texas	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Utah	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Vermont	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Virginia	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Washington	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Washington DC	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
West Virginia	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24
Wisconsin	1221	219	18.0	16.2	19.8		1000	14	1.4	0.9	2.0		1000	24
Wyoming	1221	801	65.6	63.2	68.0		1000	24	2.4	1.7	3.1		1000	24

- **MMR in the US 1987-2020:** Checked for missing values: No missing values found.
- **Verified data types:** Correct types for numeric and categorical data.
- **Outlier detection:** Identified high mortality rates for further analysis.

```
print("\nMissing Values:")
print(missing_values)

print("\nSummary Statistics:")
print(summary_statistics)

print("\nOutlier Check (Negative Values):")
print(outlier_check)
```

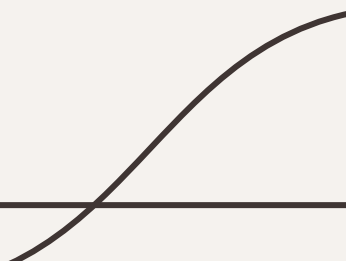
```
Data Info:
None

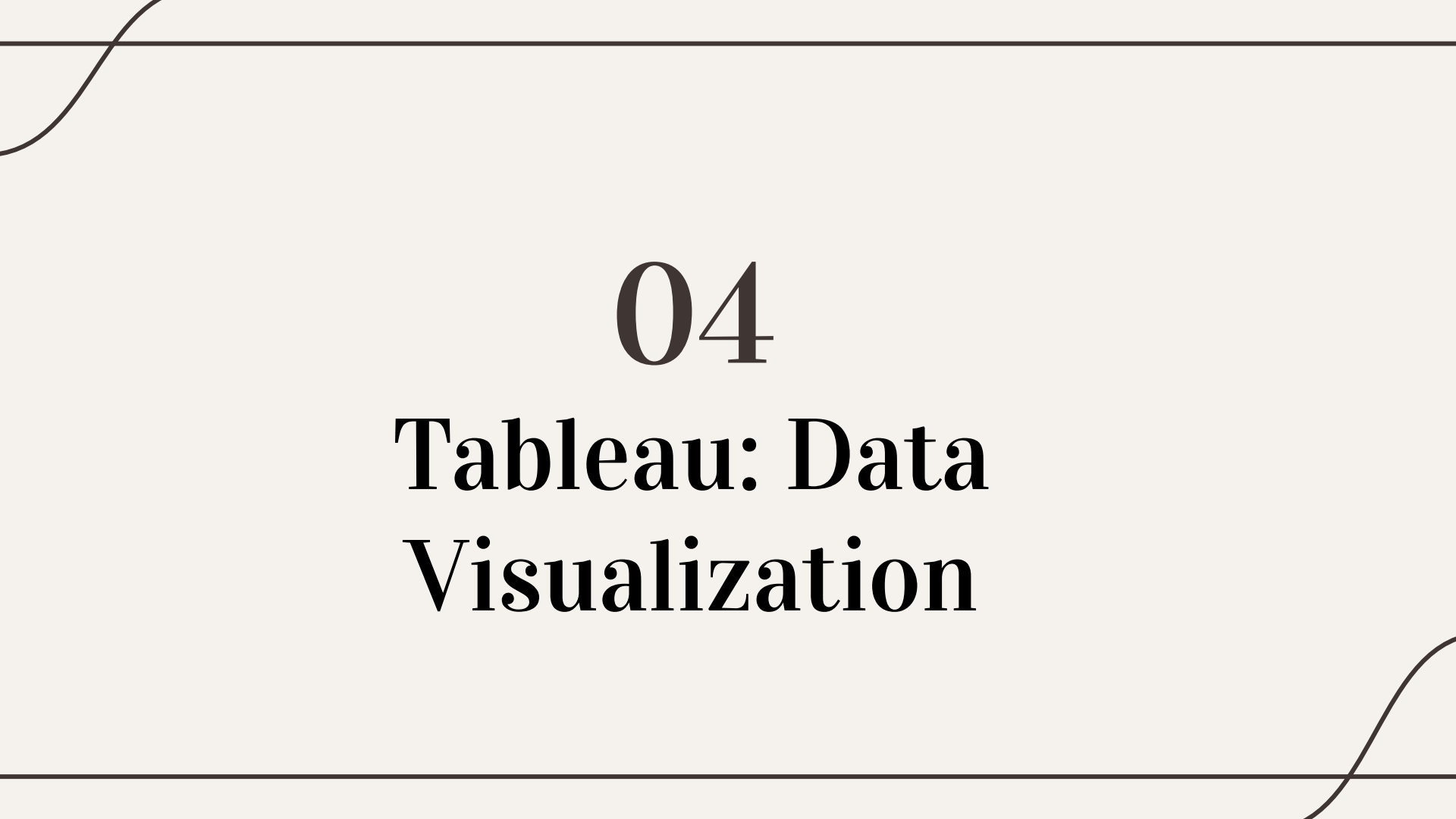
Missing Values:
Data As Of          0
Jurisdiction         0
Group               0
Subgroup            0
Year of Death       0
Month of Death      0
Time Period         0
Month Ending Date   0
Maternal Deaths    0
Live Births         0
Maternal Mortality Rate 0
dtype: int64
```

Data Wrangling

- Excel
- Python

Data Visualization

- Excel
 - Python
 - Tableau
- 



04

Tableau: Data Visualization



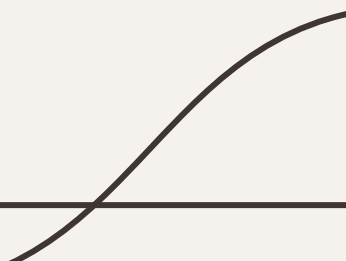
05

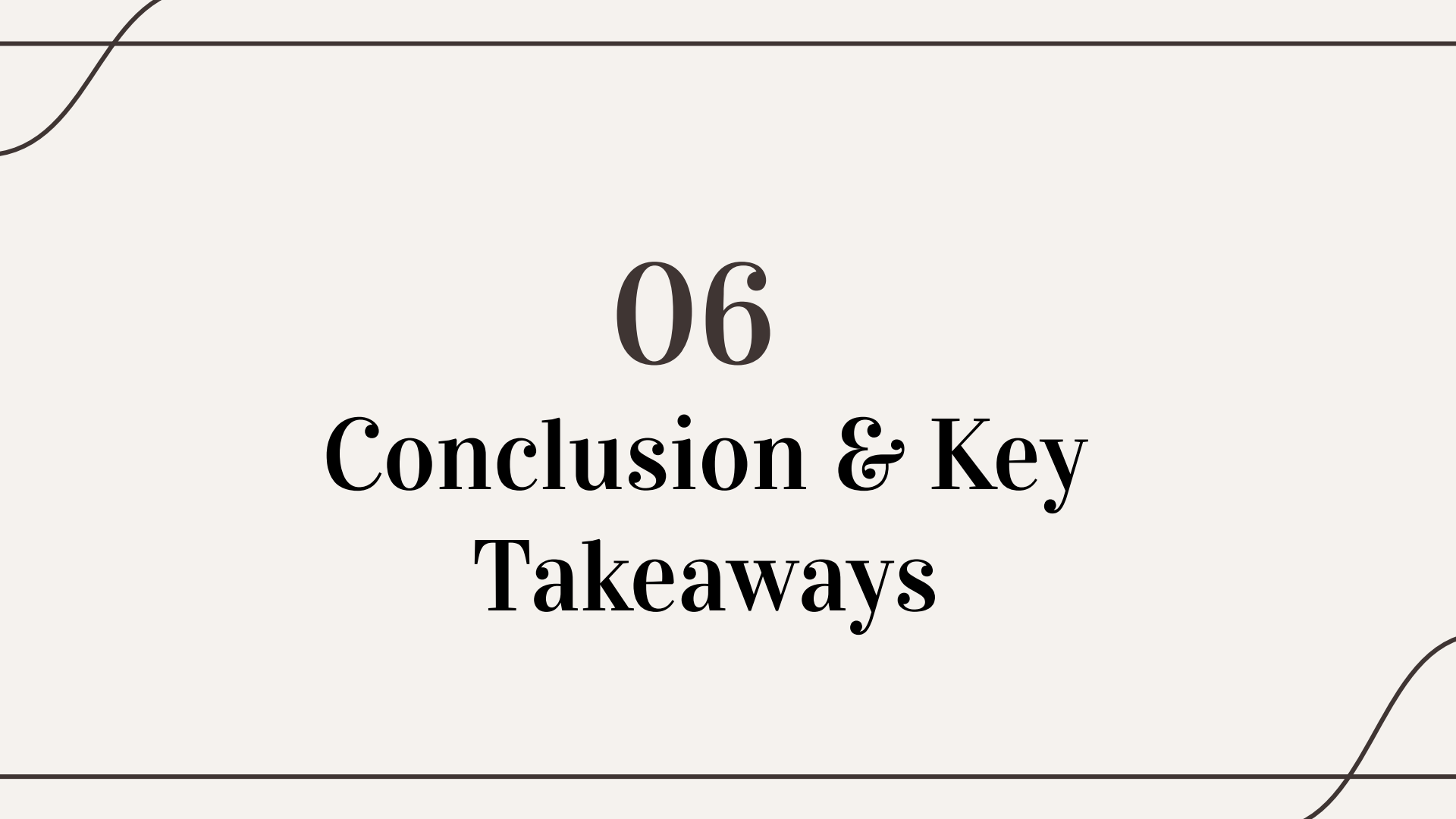
Challenges & Learnings

Challenges

1. Data Availability and Quality
 - a. Difficulty in accessing comprehensive datasets for recent years.
 - b. Variability in data formats across sources required significant organizing.
2. Data Integration
 - a. Merging datasets like PRAMS, CDC VSRR, and Pregnancy Mortality Surveillance Systems was complex.
 - b. Addressing inconsistencies in variable definitions and data points.
3. Visualization and Insights
 - a. Our team had limited experience with Tableau, which added to the learning curve in creating effective and interactive dashboards.
 - b. Limited availability of real-time data for post-pandemic trends.
4. Socioeconomic and Demographic Variations
 - a. Challenges in interpreting the interplay between socioeconomic factors and maternal mortality rates.
 - b. Balancing statistical analysis with the narrative to highlight disparities.

Learnings

1. Importance of Data Preparation
 - a. Thorough data preparation is essential for generating meaningful insights.
 2. Collaboration
 - a. Team brainstorming led to deeper insights and creative solutions.
 3. Learning New Tools
 - a. Gaining hands-on experience with Tableau enhanced our data visualization skills.
 4. Context Matters
 - a. Socioeconomic and demographic factors provide critical context for understanding maternal health challenges.
- 



06

Conclusion & Key
Takeaways

Conclusion

1. **Maternal Mortality Trends**

- Maternal mortality rates increased during and post-COVID-19 (until 2023) compared to pre-pandemic years.
- Clear disparities are evident across demographic and socioeconomic groups.

2. **Healthcare System Insights**

- Despite high spending, gaps in access to quality maternal care (insurance) persist in the U.S.
- Preventable causes remain a significant contributor to maternal deaths.

3. **Need for Targeted Interventions**

- Policies must address socioeconomic inequalities and improve access to care.
- Investments in maternal health infrastructure are critical.

Key Takeaways

1. **Data-Driven Decisions**

- Reliable and comprehensive data is essential for actionable insights into maternal health.

2. **Equity Focus**

- Efforts must prioritize underserved communities to reduce disparities in maternal outcomes.

3. **Call to Action**

- Strengthen maternal healthcare policies and ensure consistent, high-quality care nationwide.
- Enhance data collection and real-time analysis for proactive decision-making.